

Concepts of systems, models, and system architectures; n-tier architectures, distributed computing; cloud computing

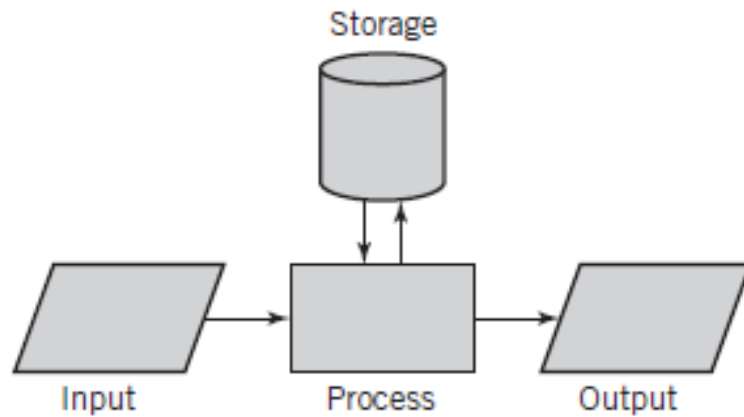
This presentation contains images that are part of the textbook we use in this course: Englander, Irv, and Wilson Wong. *The architecture of computer hardware, systems software, and networking: An information technology approach*. John Wiley & Sons, 2021, and the goal of this presentation is for pedagogical purposes ONLY and no ownership is claimed, by its author, Bernal Manzanilla Saavedra, on any of the images nor in the contents of the book.

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Thursday 14th Sept 2023
Bernal Manzanilla Saavedra

The basic Architecture of Computers

A Computer Process



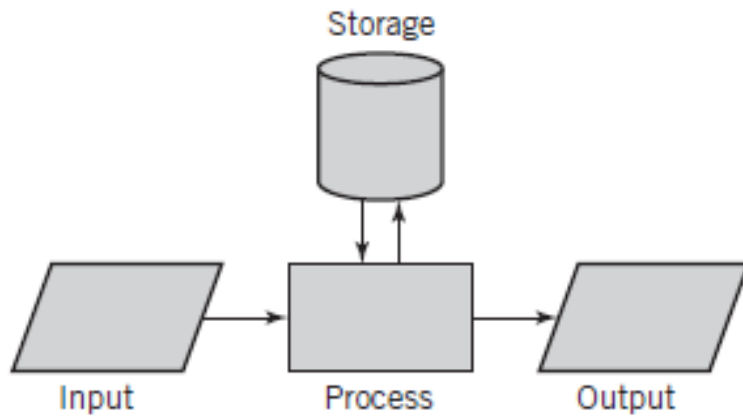
Englander, Irv, and Wilson Wong. *The architecture of computer hardware, systems software, and networking: An information technology approach*. John Wiley & Sons, 2021.



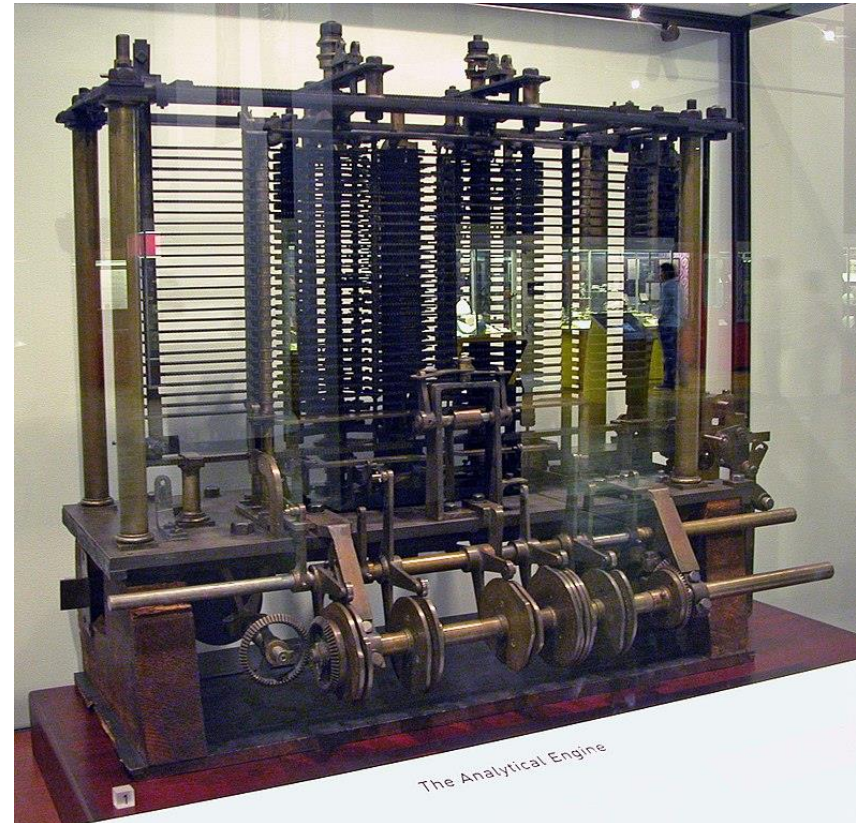
<https://www.nms.ac.uk/explore-our-collections/stories/science-and-technology/jacquard-loom/>

The basic Architecture of Computers

A Computer Process



Englander, Irv, and Wilson Wong. *The architecture of computer hardware, systems software, and networking: An information technology approach*. John Wiley & Sons, 2021.

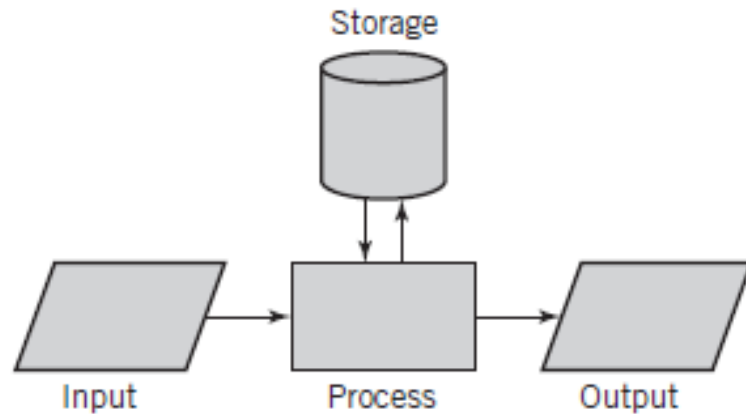


Analytical Engine, trial model, Science Museum Group

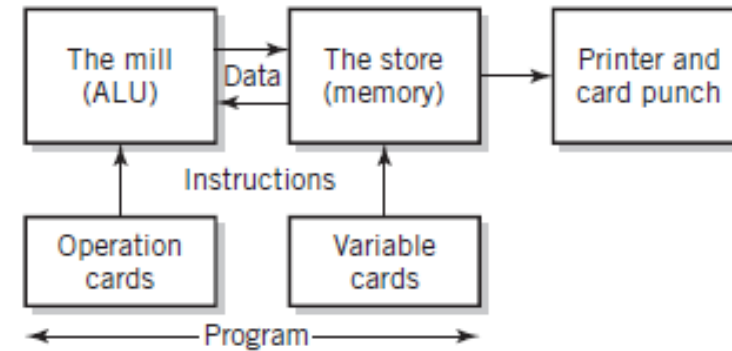
<https://blogs.bodleian.ox.ac.uk/adalovelace/2018/07/26/adalovelace-and-the-analytical-engine/>

The basic Architecture of Computers

A Computer Process



Block Diagram of Babbage's Analytical Engine



Source: Computer Architecture and Organization, 2e., J. Hayes, copyright 1988 by McGraw-Hill Companies p. 14.

The basic Architecture of Computers

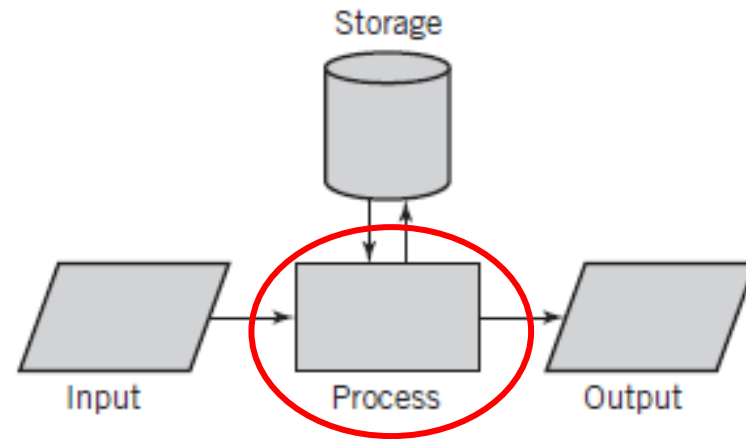
- 1.** The computer hardware, which provides the physical mechanisms to input and output data, to manipulate and process data, and to electronically control the various input, output, and storage components.
- 2.** The software, both application and system, which provides instructions that tell the hardware exactly what tasks are to be performed and in what order.
- 3.** The data that is being manipulated and processed. This data may be numeric, it may be alphanumeric, it may be graphic, or it may take some other form, but in all cases it must be representable in a form that the computer can manipulate.

The basic Architecture of Computers: CPU

Conceptually, a Central Processing Unit (CPU) itself is often viewed as a composition of three primary subunits:

1. The **arithmetic/logic unit (ALU)** where arithmetic and Boolean logical calculations are performed.
2. The **control unit (CU)**, which controls the processing of instructions and the movement of internal CPU data from one part of the CPU to another.
3. The **interface unit**, interconnects the CPU with memory and also with the various I/O modules.

A Computer Process



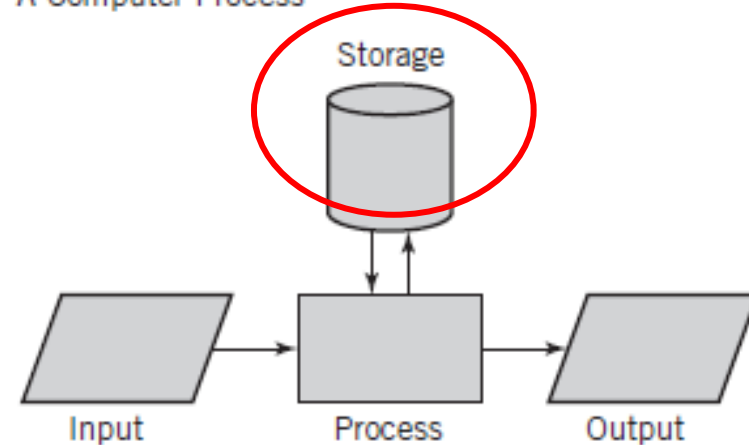
The basic Architecture of Computers: Storage

The main memory, often known as primary storage, working storage, or **RAM**(for **r**andom **a**ccess **m**emory), holds programs and data for access by the CPU.

Primary storage is made up of a large number of cells, each numbered and individually addressable. Each cell holds a single binary number representing part of a data value or part of an instruction.

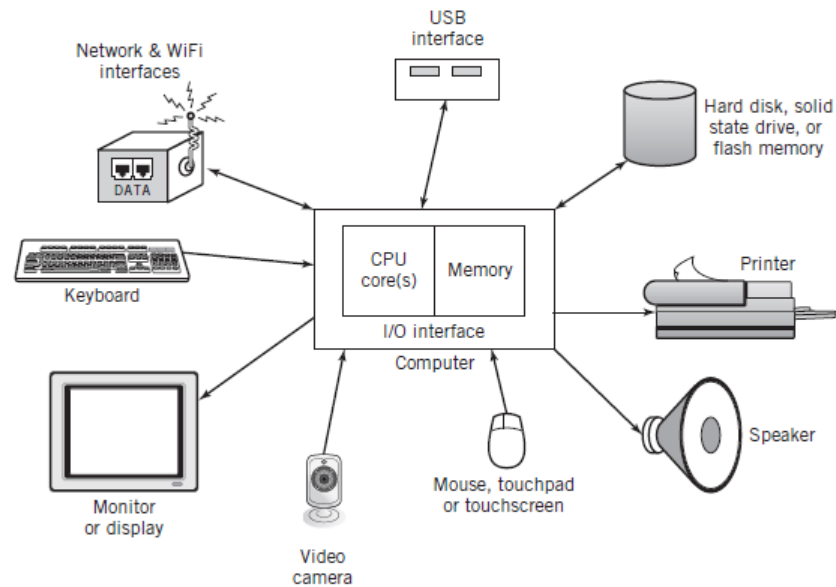
The smallest addressable size of the cell in most current computers is 8 bits, known as a **byte** of memory

A Computer Process

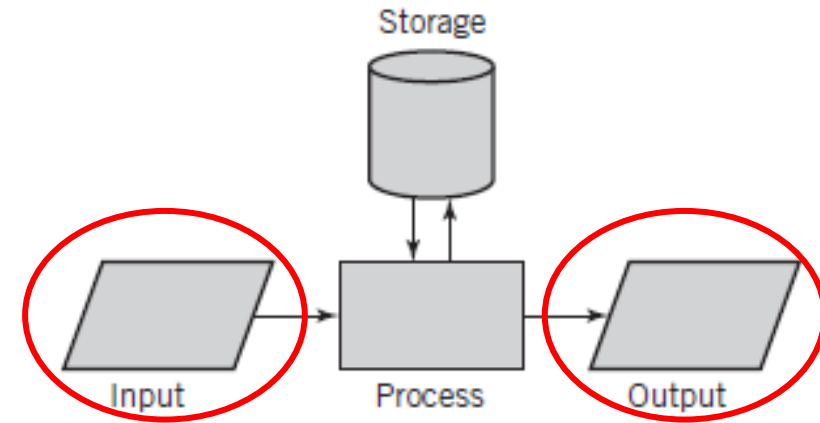


The basic Architecture of Computers: I/O

A Typical Personal Computer System



A Computer Process



Software

There are two major categories of software: system software and application software

- **System software** helps you to **manage** your **files**, to load and execute programs (**applications**), and executes your commands. The system software programs that manage the computer are collectively known as an **operating system**
- **Application (App) software: Gove me some examples!**

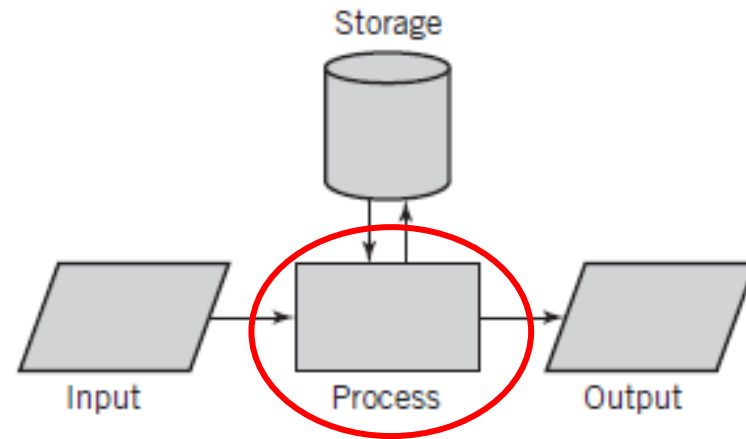
Hint: think what software you use when you do your homework...

System Software

The basic Architecture of Computers: OS

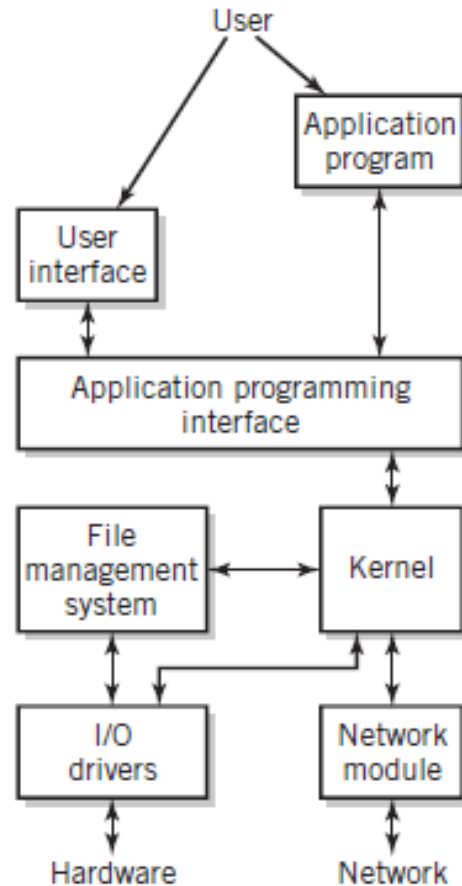
- The operating system's **application programming interface (API)**, acts as an interface for application programs and utilities to access the internal services provided by the operating system.
- **Kernel** memory addressing and allocation, scheduling for each application, communication between programs, manages and arranges services provided by other modules, security.
- **File management system** allocates and manages secondary storage space and translates file requests from their name-based form into specific I/O requests.

A Computer Process

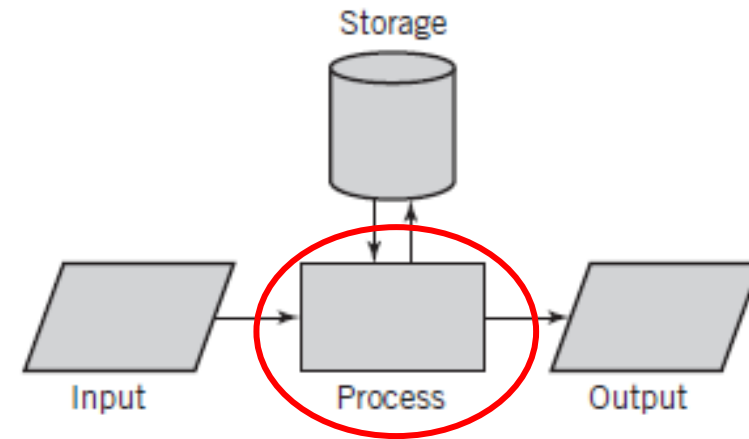


The basic Architecture of Computers: OS

Simplified Operating System
Block Diagram



A Computer Process



The basic Architecture of Computers: OS

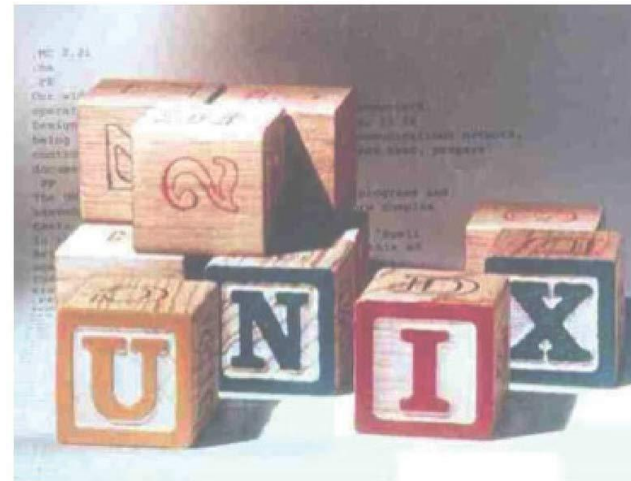
- First Operating system was developed by General Motors Research Laboratories in 1953–1954 for their IBM 701 computer
- 1963, Burroughs released its Master Control Program (MCP). Contained high-level language facilities and support for multiprocessing (with two identical CPUs). Supported virtual storage, as well as powerful multitasking capabilities
- IBM introduced OS/360 as the operating system for its new System/360 in 1964.
- In 1962, a group at MIT known as Project MAC introduced the concept of time-sharing with an experimental operating system called CTSS. MIT, Bell Labs, and GE formed a partnership to develop a time-sharing system. The system was called MULTICS (Multiplexed Information and Computing Service)

Operating Systems: UNIX

When Bell Labs withdrew from the MULTICS project, Ken Thompson, a MULTICS researcher, turned to the development of a small personal operating system, which he called Unics, later UNIX, to contrast it from MULTICS. He was later joined by Dennis Ritchie. The original UNIX development was performed on a Digital PDP-7 minicomputer and later moved to a PDP-11 minicomputer, then to the Digital VAX computer. These were popular computer systems supplied by the Digital Equipment Corporation between 1964 and 1992. Originally, the system was written in assembly language, but Ritchie developed a new high-level language, which he called C, and the operating system was largely rewritten in C.

UNIX A History and a Memoir

Brian Kernighan



SECOND EDITION

THE



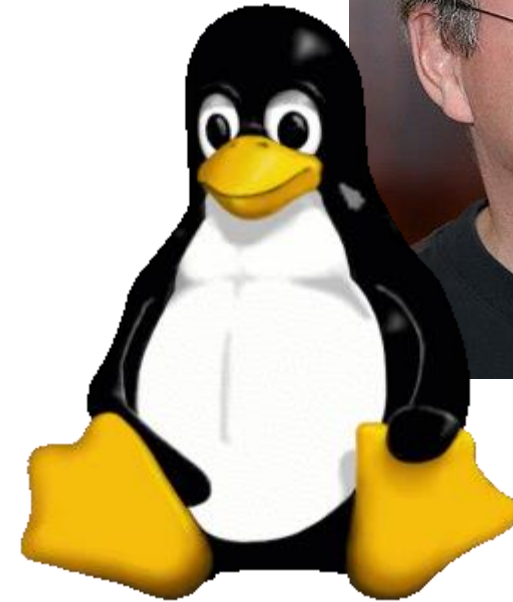
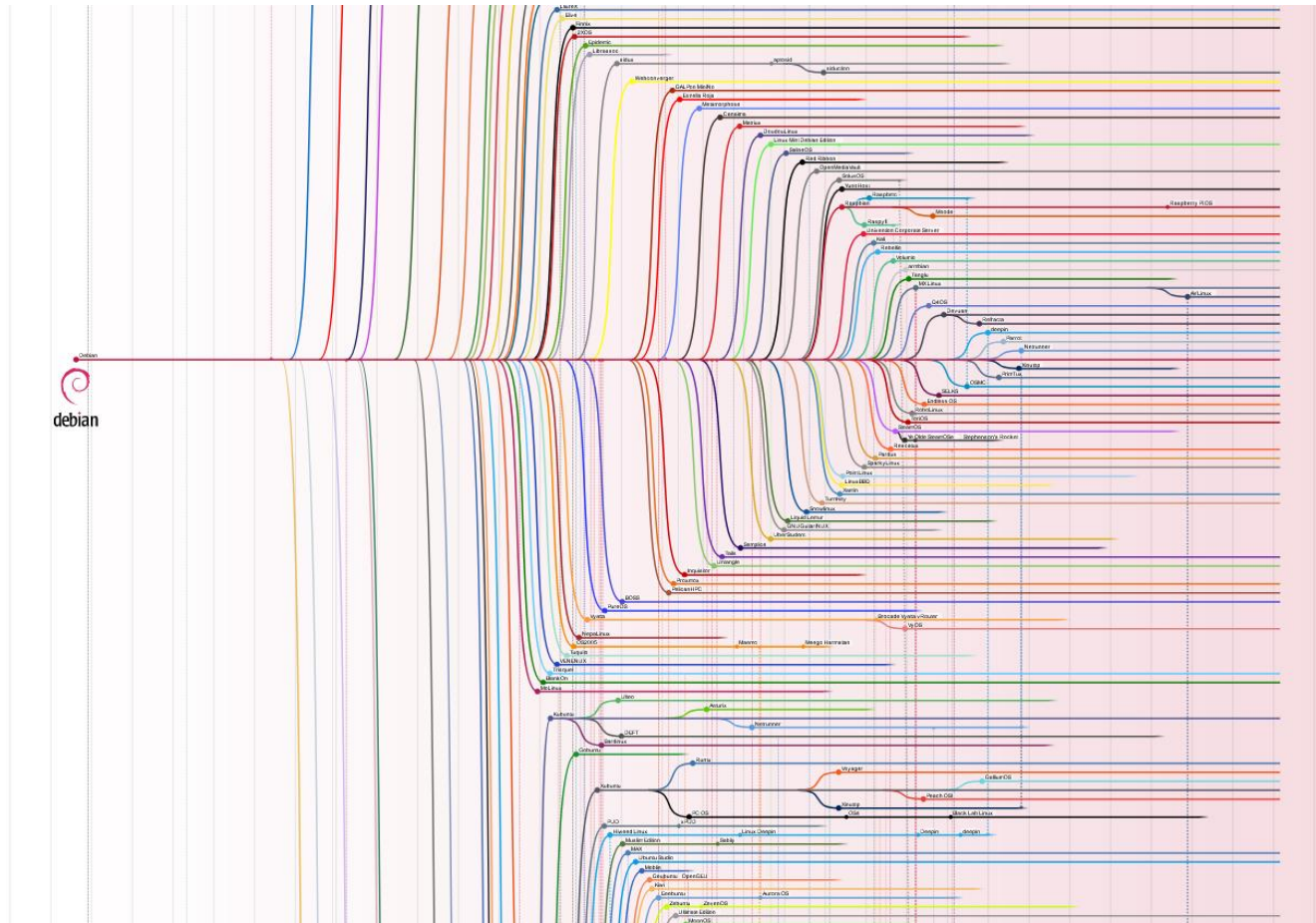
PROGRAMMING
LANGUAGE

BRIAN W. KERNIGHAN
DENNIS M. RITCHIE

PRENTICE HALL SOFTWARE SERIES

Operating Systems: LINUX and UNIX

Linux kernel is a [free and open-source](#),^{[12]:4} [monolithic](#), [modular](#), [multitasking](#), [Unix-like](#) operating system [kernel](#). It was originally written in 1991 by [Linus Torvalds](#) for his [i386](#)-based PC, and it was soon adopted as the kernel for the [GNU operating system](#), which was written to be a [free \(libre\)](#) replacement for [Unix](#).



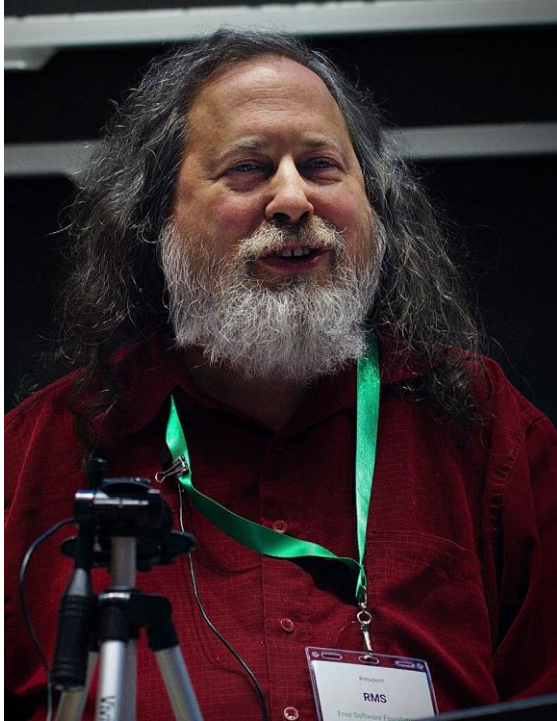
https://en.wikipedia.org/wiki/Linux_kernel

and a must visit!

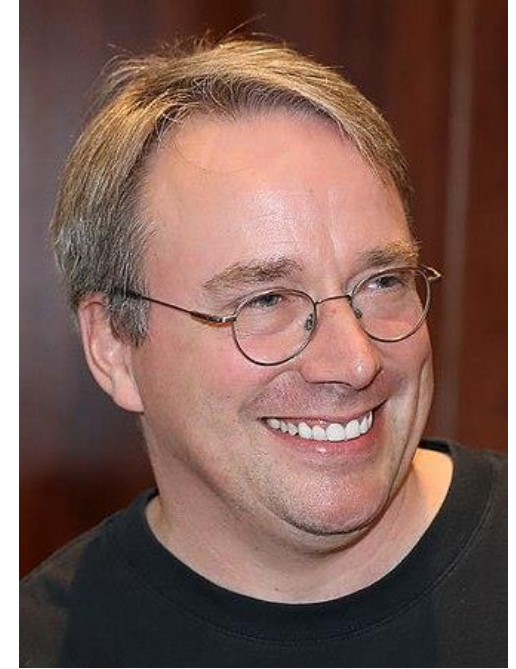
https://en.wikipedia.org/wiki/List_of_Linux_distributions

Operating Systems: LINUX and UNIX

GNU foundation and the Open-source license: Freedom and education for all!!



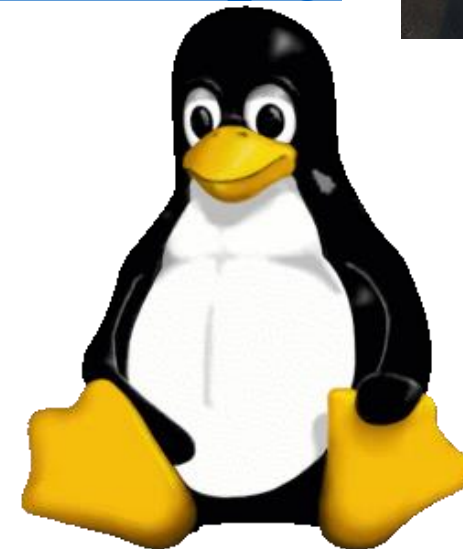
https://dragonball.fandom.com/wiki/Fusion_Saga



GNU

<https://www.gnu.org/home.en.html>

https://en.wikipedia.org/wiki/GNU_Project



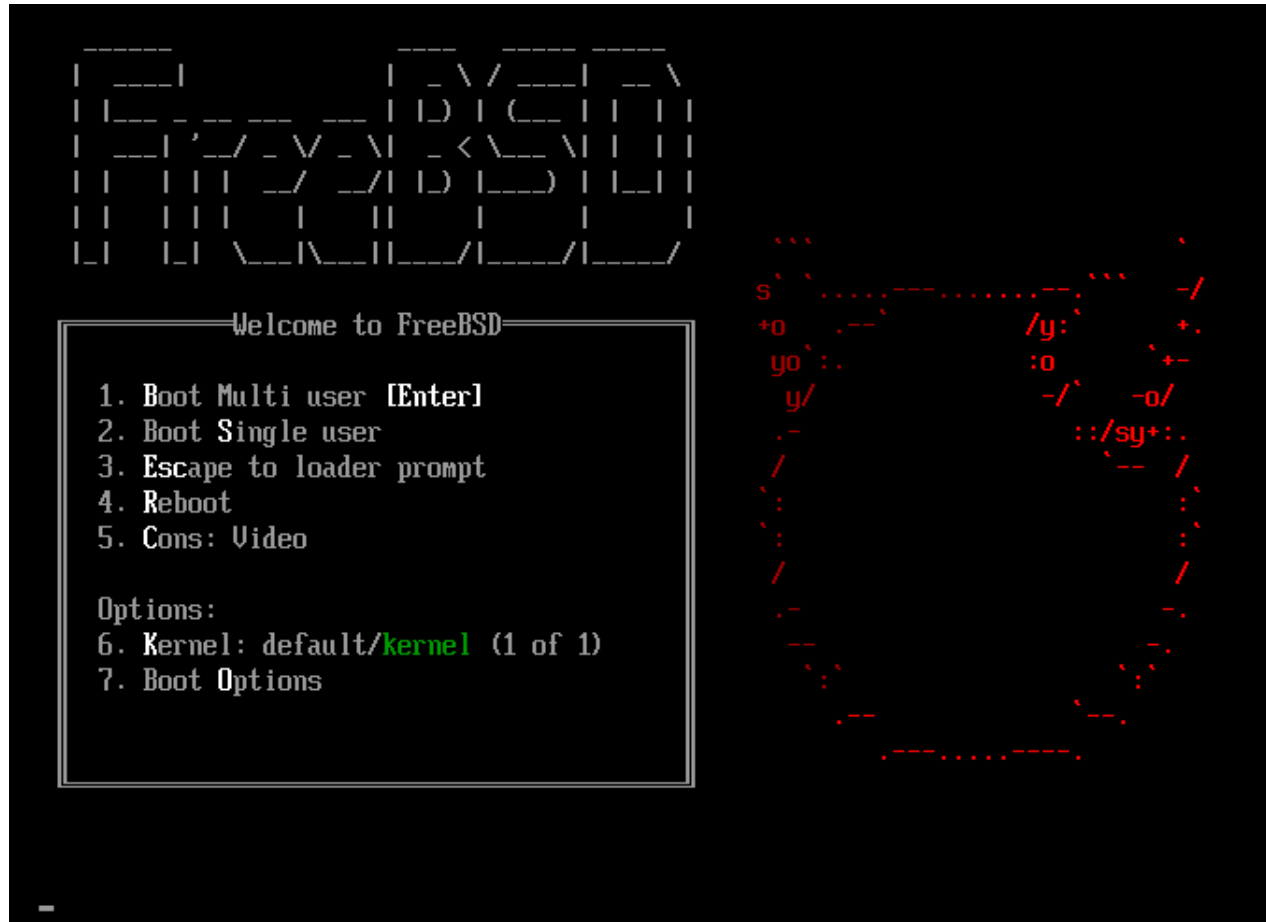
Tux

https://en.wikipedia.org/wiki/Linux_kernel

and a must visit!

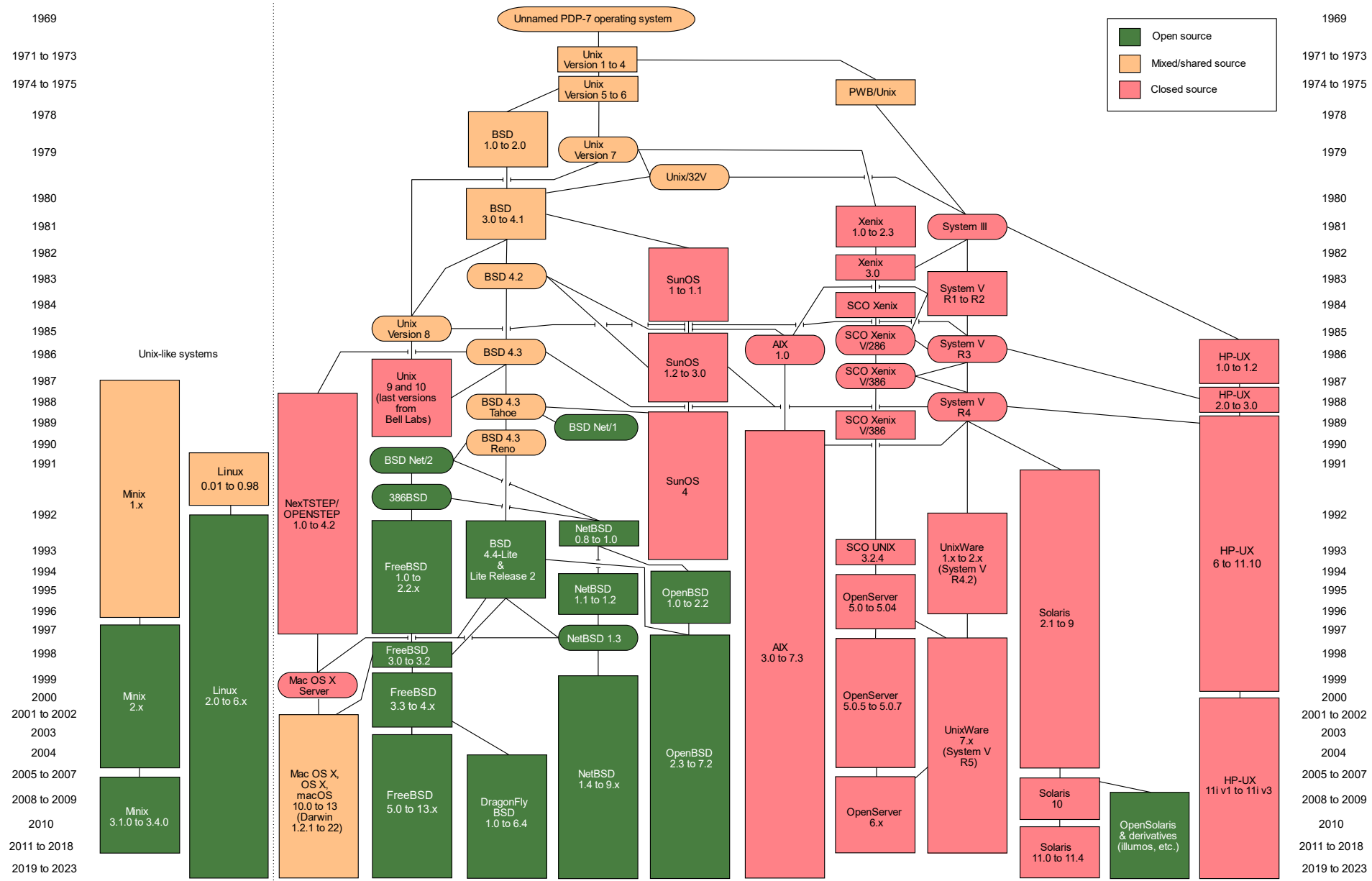
https://en.wikipedia.org/wiki/List_of_Linux_distributions

Operating Systems: Berkeley Software Distribution



FreeBSD has similarities with [Linux](#), with two major differences in scope and licensing: FreeBSD maintains a complete system, i.e. the project delivers a [kernel](#), [device drivers](#), [userland](#) utilities, and [documentation](#), as opposed to Linux only delivering a [kernel](#) and drivers, and relying on third-parties for system software;^[5] FreeBSD [source code](#) is generally released under a [permissive BSD license](#), as opposed to the [copyleft GPL](#) used by Linux

Operating Systems: LINUX and UNIX

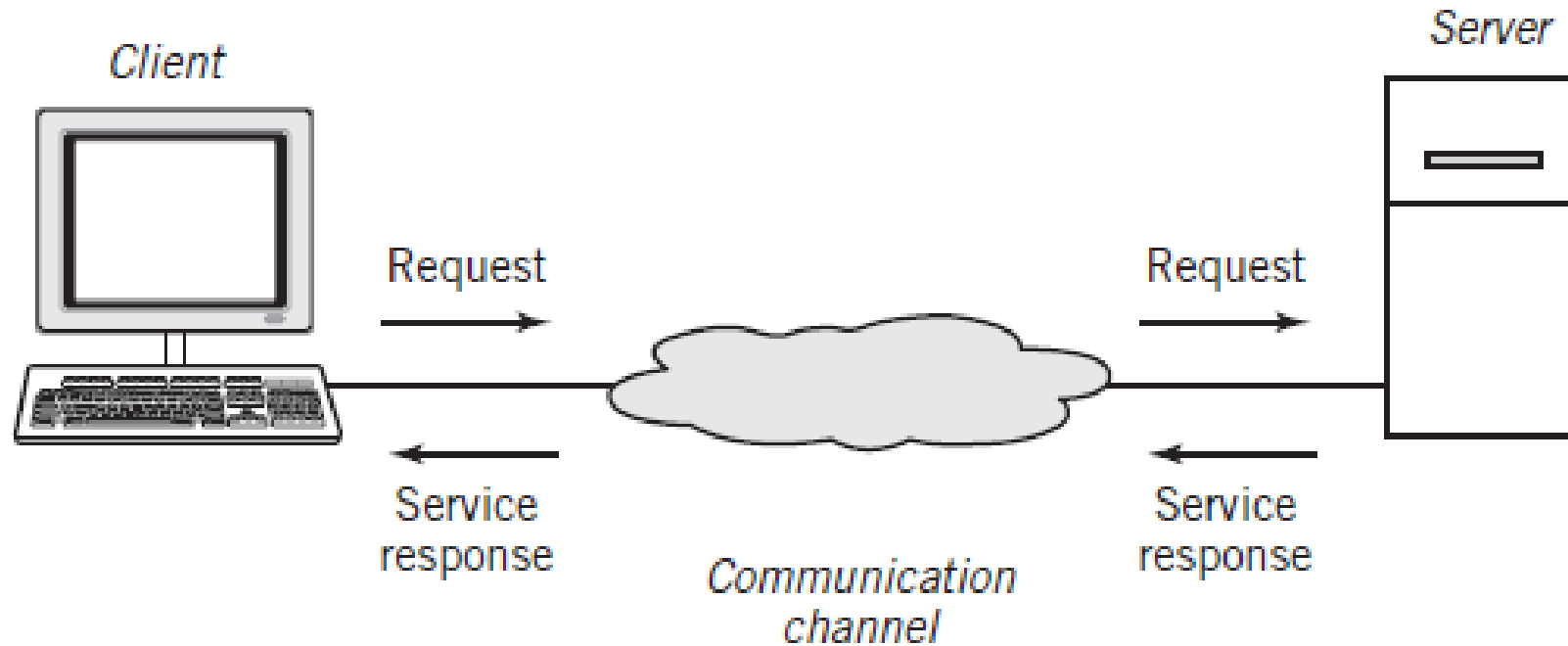


Task breakdown and message passing, not faster sequential calculation

- It was in this context that new architecture as the modern iX Intel processor were born, distributing tasks into many core (processors) instead of aiming at faster sequential execution.
- As switching and transmission technology evolved, and bandwidth increased, the latency times for computers to communicate decreased.
- Why not connecting many computers? Internet

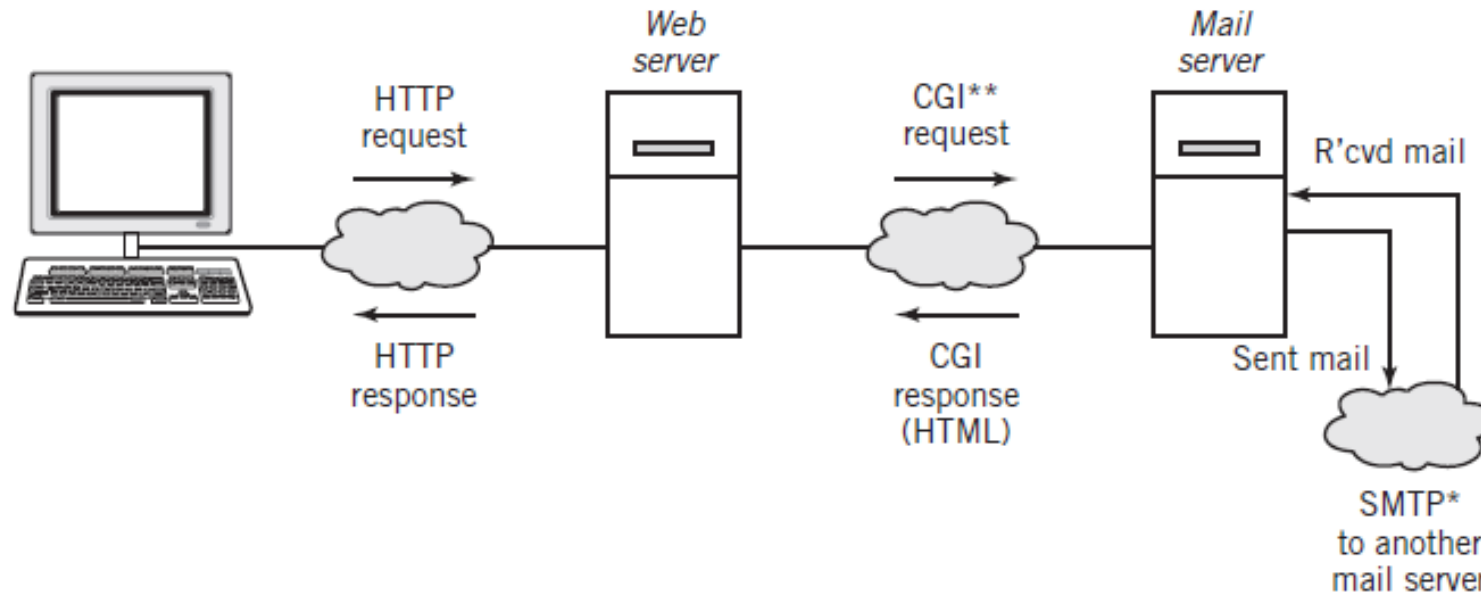
2-Tier architecture

Basic Client–Server Architecture



3-Tier architecture

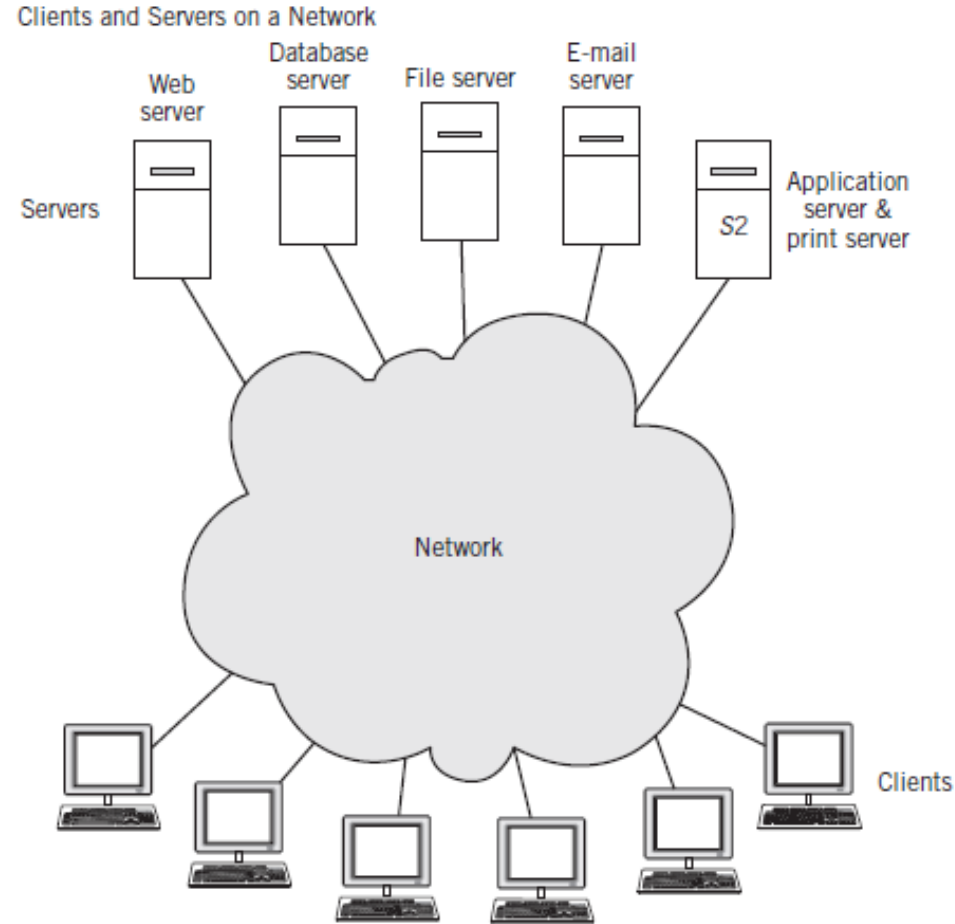
Three-Tier Web-Based E-Mail Architecture



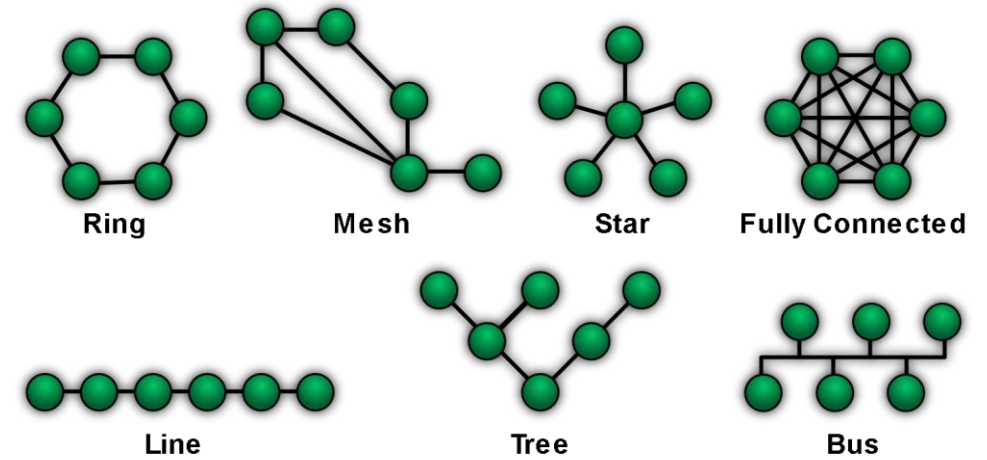
*SMTP: Simple Mail Transfer Protocol

**CGI: Common Gateway Interface

n-Tier architecture



Network topologies



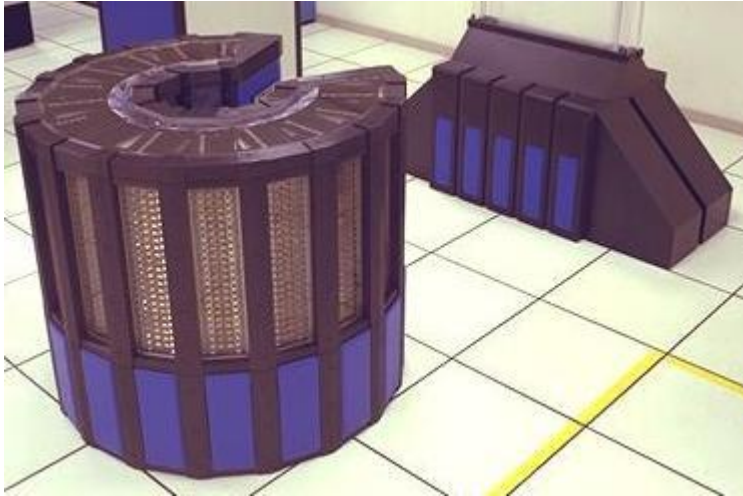
https://en.wikipedia.org/wiki/Network_topology#/media/File:NetworkTopologies.svg

Englander, Irv, and Wilson Wong. *The architecture of computer hardware, systems software, and networking: An information technology approach*. John Wiley & Sons, 2021.

Departing from Moore's Law

- As chip were miniaturized costs tended to decrease.
- We hit a wall around 2004. Chip miniaturization decelerated.
- Multi core (many sequential CPU's) were the new paradigm. Parallelism and NOT clock speed became the path to faster computing
- Along with miniaturization transmission speed and bandwidth of signals increased.
- The predecessor of the modern Internet began, i.e. the **Advanced Research Projects Agency Network ([ARPANET](#))**, in 1969. Decommissioned in 1989.
First network to use packet switching and TCP/IP
- Just like cores, why not connecting many computers and making them work in a synchronous manner?
Enter the Cluster as the new Supercomputer.

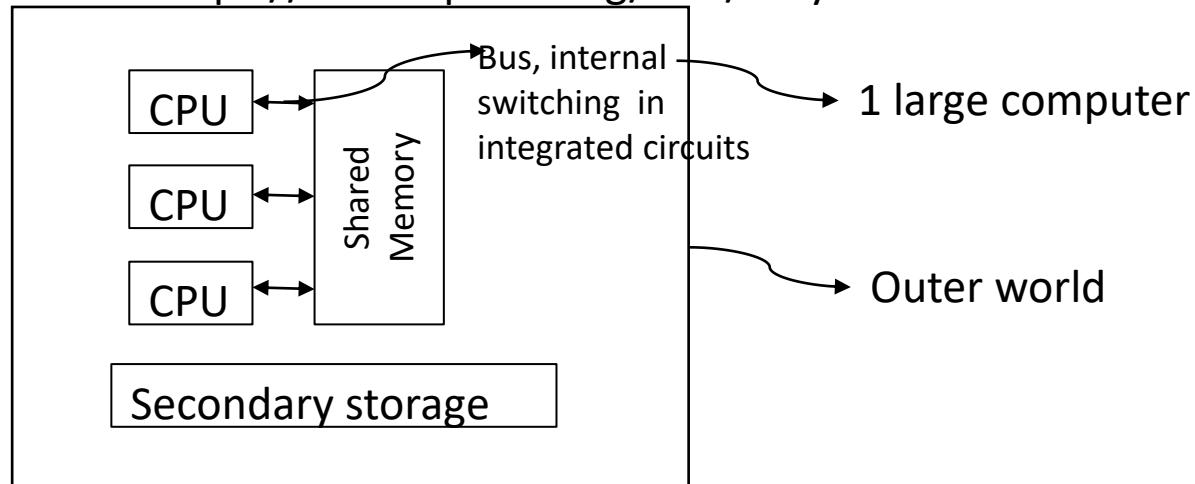
Sequential versus Parallelism



<https://en.wikipedia.org/wiki/Cray>

1 large computer

Bus, internal
switching in
integrated circuits

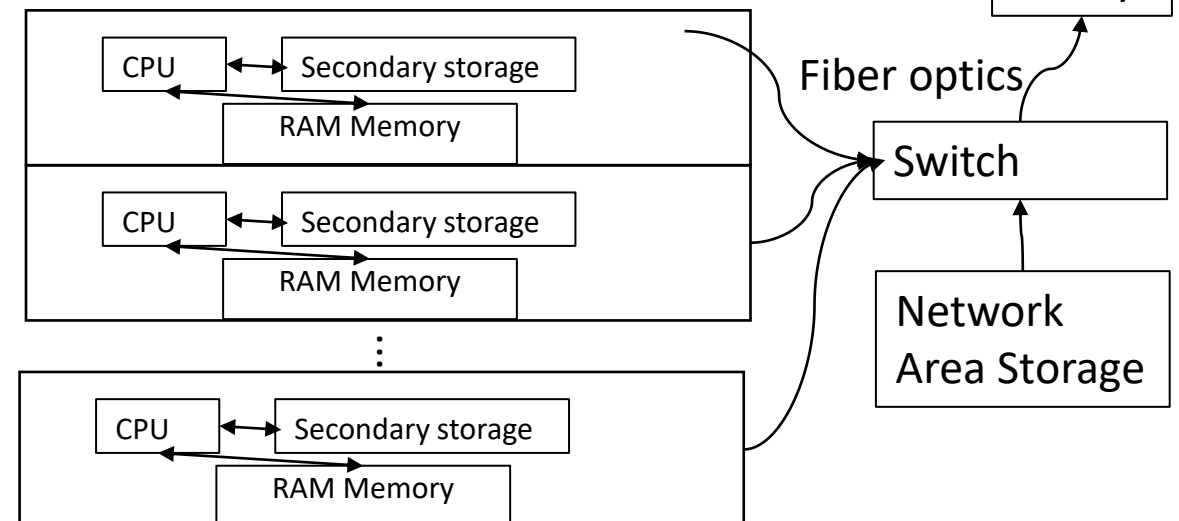


<https://www.hpe.com/us/en/compute/hpc.html>



"n" small
computers

Outer
world



Distributed Computing

Since computers all operate so similarly, regardless of size or type, it is not difficult today to transfer data between these different systems, allowing each system to do part of the processing for higher overall efficiency. This concept is known as **distributed computing**. The fact that different types of computers can work together, share files, and communicate successfully is known as **open computing**

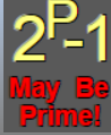
Berkeley SETI Research Center: SETI@home



<https://www.warnerbros.com/movies/contact>

<https://setiathome.berkeley.edu/>

Distributed Computing



Great Internet Mersenne Prime Search

GIMPS

Finding World Record Primes Since 1996



Username

Password

Log In

Forgot password

Home

Get Started

Current Progress

Create Account

Reports

Manual Testing

More Information / Help

Donate
Make a donation

Welcome to the Great Internet Mersenne Prime Search

Known Primes

GIMPS Milestones

Recent Results

Recent Cleared

Work Distribution Map

Active Assignments

Follow these instructions

Quick Links: Download Primes Progress Overview Milestones History

All exponents below 54 888 729 have been tested and verified.
All exponents below 113 246 459 have been tested at least once.

Today's Numbers

Teams	1 572
Users	252 549
CPUs	2 618 439
GFLOP/s	4 820 768
GHz-Days	2 410 384

Previous Day Stats

First Prime Tests	206
Verified Prime Tests	341
Newly Factored	57

2022-Aug-09

Prime95 version 30.8 released

Version 30.8 is now available with a few minor enhancements. P-1 stage 2 is now much faster if you can give prime95 lots of memory to use. This is not a required upgrade. Should you decide to upgrade, if any workers are currently in P-1 stage 2 wait for P-1 to finish before upgrading. If you have any upgrade questions, ask in [this thread](#) at Mersenne Forum.

2021-Nov-28

Prime95 version 30.7 released

Version 30.7 is now available with a few minor enhancements. P-1 stage 2 is now faster, Intel Alder Lake CPU is supported. This is not a required upgrade. Should you decide to upgrade, if any workers are currently in P-1 stage 2 wait for P-1 to finish before upgrading. If you have any upgrade questions, ask in [this thread](#) at Mersenne Forum.

2021-Oct-06

All tests smaller than the 48th Mersenne Prime, M(57 885 161), have been verified

M(57 885 161) was discovered eight and half years ago. Now, thanks to the largely unheralded and dedicated efforts of thousands of GIMPS volunteers, every smaller Mersenne number has been successfully double-checked. Thus, M(57 885 161) officially becomes the 48th Mersenne prime. This is a significant milestone for the GIMPS project.

2021-Apr-08

First-time Lucas-Lehmer Testing Ends

One year ago, first-time PRP primality testing with proofs was introduced. It has been a huge success, saving GIMPS tens of thousands future double-checks. Going forward, the server will no longer make available exponents for first time Lucas-Lehmer tests. Users that have not yet upgraded to prime95 version 30.3 or gpuowl for GPUs should do so. Failure to upgrade will result in necessary

<https://en.wikipedia.org/wiki/Distributed.net>